Course Code: CA105B	Course Name: Advanced Java Programming	L	T	P	C
		2	0	2	3

Pre-requisite: Java OOP's

Course Objectives:

1. Apply old and Java8+ new features to build java programs.
2. Apply to use Stream API to write efficient, functional-style code for processing collections for DSA.
3. Implement JDBC for database connectivity, enabling students to perform CRUD operations and manage relational data in Java applications.
4. Testing of RESTful API with Spring Boot.

Course Outcome:

Students will be able to:		Knowledge Category	Bloom Level
CO1	Apply OOP's concepts using Java	P	3
CO2	Apply new java features to build java programs.	P	3
CO3	Examine the use of functional programming constructs in the Java Stream API for collection processing.	P	3
CO4	Use JDBC APIs to perform CRUD database operations in Java applications	P	3
CO5	Analyze web and RESTful services developed using Spring Boot and Spring Framework concepts.	P	4

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High):

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	1	2	1	2	-	-	-	1
CO2	2	2	2	2	-	-	-	2
CO3	2	2	2	2	1	-	-	3
CO4	1	2	1	2	1	-	-	1
CO5	2	2	3	3	2	-	-	3

Detailed Syllabus

UNIT 1	**Revision of OOP's core concepts**: Object-oriented programming principles, class and object, static, final, this, super, constructor, inheritance, encapsulation, polymorphism, abstraction, abstract class, interface. **Hands-on: Programs using OOP's concepts**	**9**

Functional Programming in Java		
UNIT 2	**Java New Features:** Functional Interfaces, Lambda Expression, Method References, Stream API, Default Methods, Static Method, Private Method, For Each Method, Java Module System, Local Variable Type Inference, Switch Expressions, Yield Keyword, Text Blocks, Records, Sealed Classes. **Hands-on:** Programs using Lambda	**9**
Stream API in Java		
UNIT 3	**Stream API:** Creating Streams, use of filter(), map(), reduce(), sorted(), sorted(Comparator), forEach(), toArray(), collect(), distinct(), limit(), skip(), count(), max() and min() method. **Hands-on:** Find out list of pass and fail students in given list those attendance is more than 60%.	**9**
JDBC		
UNIT 4	**JDBC:** Introduction, JDBC Architecture, Types of Drivers, Statement, ResultSet, Read Only ResultSet, Updatable ResultSet, Forward Only ResultSet, Scrollable ResultSet, PreparedStatement, Connection Modes, SavePoint, Batch Updations, CallableStatement, BLOB & CLOB**.** **Hands-on:** CRUD App using JDBC	**9**
Introduction to Spring Boot		
UNIT 5	**Spring:** Spring Core Basics-Spring Dependency Injection concepts **Spring Boot:** Spring Boot- Spring Boot Configuration, Spring Boot Annotations, BUILDING RESTFUL WEB SERVICES, Rest Controller, Request Mapping, Request Body, Path Variable, Request Parameter, GET, POST, PUT, DELETE APIs, Build Web Applications **Hands-on:** CRUD API using Spring Boot	**9**
	Total Lecture Hours	**45**

Textbook:

1. Deitel & Deitel, "Java How to program", Prentice Hall, 4th Edition, 2000.
2. Modern Java in Action – Raoul-Gabriel Urma, Mario Fusco, Alan Mycroft, 3rd Edition
3. Java: The Complete Reference, Eleventh Edition, 11th Edition. by Herbert Schildt. Released, 2018.
4. Craig Walls, "Spring Boot in Action", 5th Edition, Manning Publication, 2015.

Reference Books:

1. Steven Holzner, "Java Black Book", Edition, Dreamtech, 2001.
2. Balagurusamy E, "Programming in Java", 5th Edition, McGraw Hill, 2017.
3. Java: A Beginner's Guide by Herbert Schildt, 9th Edition, Oracle Press, 2022.
4. Greg L. Turnquist "Learning Spring Boot 2.0 – 2nd Edition", Packt Publication, 2017.
5. AJ Henley Jr (Author), Dave Wolf, "Introduction to Java Spring Boot, Learning by Coding", Independently Published, 2019.

Mode of Evaluation:

Evaluation Scheme						
MSE		**CA**			**ESE**	**Total Marks**
MSE1	**MSE2**	**CA1**	**CA2**	**CA3**	75	150
30	30	6	6	3		
60		15				